

SwiftDrop Analytics: Architecting a Single Source of Truth for E-Commerce Profitability

This project, **SwiftDrop Analytics**, is an end-to-end data pipeline and business intelligence solution designed to optimize the operations and profitability of a delivery-based e-commerce platform.

Project Summary

Situation

SwiftDrop, a growing e-commerce and delivery service, was struggling to gain a clear view of its financial health and operational efficiency. With data spread across thousands of order, product, and delivery records, the business faced unknown revenue leaks, potentially high return rates, and inconsistent delivery performance across different warehouses.

Task

My goal was to build a robust analytics pipeline to answer critical business questions:

- What is our actual gross profit after accounting for discounts, Cost of Goods Sold (COGS), and shipping?
- Which warehouses are underperforming in terms of delivery speed?
- How do discounts impact our return rates?
- Which products are "low-margin" and threatening our bottom line?

Action

I developed a multi-stage solution using **SQL, Python, and Power BI**:

- **Data Ingestion & Cleaning:** I used **MySQL** to ingest raw CSV data. I performed extensive cleaning, including standardizing date formats, trimming whitespace from status fields, and using **Regular Expressions (Regex)** to extract measurement units (e.g., "kg", "ml") from product names to create new feature columns.
- **Advanced Analysis:** Using **Python (Pandas)** and **SQL**, I calculated complex metrics such as **Average Order Value (AOV)** per customer and **Profit Margin %** by category.
- **Techniques:** I utilized **Joins and Aggregations** to merge disparate datasets, **Common Table Expressions (CTEs)** for readable logic, and **Hypothesis-driven queries** to test if higher discounts led to higher returns.
- **Visualization:** I designed an interactive **Power BI Dashboard** to translate these technical findings into visual trends for stakeholders.

Result

The project successfully identified:

- A **Total Net Revenue of \$7.13M** and an **Absolute Gross Profit of \$2.25M**.

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- Specific "Low-Margin" products where the base profit margin was **less than 20%**, allowing the business to reconsider pricing or sourcing.
- The exact revenue lost to cancellations and returns, providing a clear target for operational improvement.

Detailed Project Breakdown

Feature	Description
Business Problem	A business can sell millions of dollars in products, but if shipping costs and returns are high, they could actually be losing money. This project provides the visibility needed to stop "leaking" cash and start optimizing for growth.
Real-World Impact	In the real world, logistics costs and return rates can make or break a delivery business. Optimizing these ensures sustainable growth and customer retention.
Idea Origin	The project was conceived to simulate a real-world "data engineering to data analysis" workflow, starting from raw, messy CSVs to a polished executive dashboard.
Data Collection	Data was collected from three core sources: orders.csv, products.csv, and deliveries.csv.
Data Cleaning	SQL: Used STR_TO_DATE for datetime objects, TRIM and REPLACE for string sanitation, and REGEXP_REPLACE for feature engineering. Python: Used .str.strip() to clean categorical labels.
Tools & Techniques	• Database: MySQL (ETL, Data Cleaning, CTEs). • Programming: Python (Pandas, SQLAlchemy) for data manipulation and AOV analysis. • Visualization: Power BI (DAX, Interactive Dashboards).

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Feature	Description
	Techniques: Hypothesis testing on "Discounts vs. Returns" and Value Density Analysis.
Specific Questions Analyzed	- Revenue trends by day. - Warehouse delivery performance. - Discount vs. Return correlation. - Value density (high-value, low-weight items).

Business Problem Solved

I solved the "Information Silo" problem. By integrating sales, inventory, and logistics data, I provided the business with a single source of truth. This allows management to stop guessing and start making decisions based on which warehouses are slowest and which products are actually profitable.